Thawing PBMCs for Single Cell Processing

Get Started

Prior to starting the experiment, samples should be picked out and analyzed to ensure criteria are met. All centrifugation steps need to be performed at 4°C. All instruments, buffers and consumables with sample contact should be pre-cooled and put on ice. All steps should be done on ice to ensure that integrity of sample is not compromised.

Before you begin:

- Prepare warm RPMI Complete Media: RPMI + 10% FBS
- Get an ice bucket ready
- Make sure to perform all cell thawing steps in a biosafety cabinet
- Set up a water bath to 37°C before getting cells from LN2 tank
- For maximum viability, perform all cell washes at room temperature
- Remember to MOVE QUICKLY from liquid nitrogen to water bath

Thawing & Resuspension of PBMCs

1. Remove PBMC cryovial from storage and immediately thaw in the water bath at 37°C for 2-3 min until there is a pea sized amount of ice left in cryovial and then bring vial into biosafety cabinet.

2. Transfer 5ml RPMI Complete into a 15ml tube. Using a wide-bore pipette or serological add 1 ml warm complete growth medium to the cryovial.

3. Add the entire contents of the cryovial (1ml cells + 1ml RPMI) dropwise (1 drop per 5 sec) to the 15ml conical tube while gently shaking the tube. (room temp.)

4. Centrifuge at 500XG for 5 min (acceleration/deceleration: 6)

5. Remove the supernatant entirely without disrupting the cell pellet.

6. Using a wide-bore pipette tip, add 1ml 1X PBS and gently pipette mix 5x.

7. Place cells on ice.

8. Determine the cell concentration using the Luna Cell Counter.

9. Transfer the cells into fixation conditions.

10. Move forward with fixation.

ScalePlex Fixation Protocol:

This protocol shows the steps for fixation of 8-12 samples. <u>50,000 to 200,000 nuclei per sample are required</u> to proceed with this protocol. DO NOT proceed if counts are lower

ScalePlex Reagent Kits

ScalePlex Fixation Reagents

Wash Buffer	Stored at -20ºC
Fixation Reagent	Stored at 4ºC
ScalePlex Oligo Plate	Stored at -20ºC
1x PBS	Stored at 4 ºC
DEPC	Stored at 4 ºC
DMSO	Stored at RT
Methanol	Stored at RT
AOPI	Stored at 4 ºC

ScalePlex Consumables

15-mL conical tubes	Stored at RT
5-mL DNA LoBind tubes	Stored at RT
1.5-mL DNA LoBind tubes	Stored at RT
0.5-mL Protein LoBind tubes	Stored at RT
96-well LoBind semi-skirted PCR plates	Stored at RT
Microseal 'B' PCR Plate Sealing Film	Stored at RT

Get Started

Prior to starting the fixation protocol take out the fixation reagent from the 4ºC and place at room temperature. Remove Wash Buffer from -20ºC, once thawed, invert Wash Buffer to ensure it is fully mixed, then place on ice. Remove ScalePlex Oligo Plate from -20ºC and place on ice. Aliquot 2 mL of ice-cold methanol to a 5mL Eppendorf Tube.

Nuclei Preparation & ScalePlex Fixation

1. Centrifuge samples at 500 xg for 5 minutes at 4°C
2. While centrifuging, calculate the fixation target concentration volume required for ScalePlex for the entire sample using the table below:

Target Cell Input per Sample for Fixation	Target Concentration	Volume required for Fixation
50,000	2,000 cells/µL	25 µL
100,000	4,000 cells/µL	25 µL
200,000	8,000 cells/µL	25 µL

Example: (*If desired concentration is 8000 cell/uL for. Divide the total concentration of your sample by 8000)*

Current Concentration (C_c) = 4x10^5 cells/mL
*Target Concertation (C_t) = 8,000 cells/*µL
Volume = X

$$V = \frac{C_c}{C_t} = \frac{4x10^5 cells/mL}{8000\ cells} = 0.05\ mL = 50\ \mu\text{L}$$

3. At this point you can start preparing the fixation reagent. Start by briefly spinning down the room temperature fixation reagent. In a chemical fume hood, add 50 µL DMSO to the Fixation Reagent Tube.
4. Vortex at high speed with intermittent brief spins to dissolve the Fixation Reagent. This may take up to several minutes. Ensure the pellet is fully dissolved before proceeding.
5. Briefly spin down the Fixation Reagent tube.
6. On ice, prepare Fixation Solution by adding the fixation reagent with DMSO to the 5mL DNA LoBind tube containing the 2mL of ice-cold methanol.
7. Briefly vortex the Fixation Solution mix and leave on ice. Use within 6 hours.
8. Once centrifugation is complete, carefully remove supernatant by pipetting without disturbing the pellet
9. Dilute nuclei with ice cold 1X PBS to the required target concentration calculated in step 2
10. Resuspend the nuclei by flicking. Proceed immediately to ScalePlex Fixation (for 8-12 samples).
11. Transfer 25uL of diluted cells to an Eppendorf twin.tec PCR LoBind plates 96 well plate on ice. Record the wells used in the PCR plate and their corresponding sample ID

12. Spin down the ScalePlex Oligo Plate by pulsing until it reaches 1000 xg at 4°C and place on ice.
13. Using a sharp tool, remove the seal from the appropriate number of ScalePlex Oligo wells to be used.
14. Transfer 5 µL of oligos from the ScalePlex Oligo Plate to each well of the PCR plate containing the nuclei samples. Record the ScalePlex Oligo used for each sample.
15. Set pipette to 25uL and gently pipette up and down five times to mix using a multi-channel or single channel pipettor
16. Incubate samples on ice for 5 minutes.
17. Store ScalePlex Oligo Plate containing unused oligos at -20°C.
18. Immediately before use, in the *chemical fume hood*, add 20 uL of DEPC to the Complete Fixation Reagent.
19. Briefly vortex the Complete Fixation Solution to mix.
20. In the hood, add 120 µL of the Complete Fixation Solution slowly to each tube containing the ScalePlex stained cells. Gently pipette up and down five times to mix.
21. Incubate on ice for 15 minutes.
22. While incubating add the following volume of Wash Buffer to the appropriate tube and keep in ice.

Number of Samples	8	9-12
Container Size	5ml LoBind tube	15 ml LoBind tube
Volume of Wash Buffer for first wash	2.4 mL	4 mL

23. After incubation, use a single-channel pipette to pool samples by transferring the cell suspension to the appropriate tube containing Wash Buffer. Same tip can be used.
24. Centrifuge the tube containing samples in Wash Buffer at 500 x g for 5 minutes at 4°C.
25. Carefully remove supernatant without disturbing the pellet, leaving ~200 µL of residual volume.
26. Resuspend the nuclei by flicking the tube and then add 800 µL ice cold Wash Buffer to the cells.
27. Add an additional 4 mL of ice-cold Wash Buffer to the nuclei.
28. Centrifuge the tube at 500 x g for 5 minutes at 4ºC.
29. Carefully remove supernatant without disturbing the pellet, leaving ~50 µL of residual volume.
30. Resuspend the nuclei by flicking the tube and then add 1 mL ice cold Wash Buffer to the cells.

31. Centrifuge the tube at 500 x g for 5 minutes at 4°C.
32. Carefully remove supernatant without disturbing the pellet, leaving ~50 µL of residual volume.

Number of Samples	8	9-12
Volume of Wash Buffer for resuspension	30 µL	50 µL
Final Volume	80 µL	100 µL

33. Resuspend the nuclei by flicking the tube and add the appropriate volume of ice-cold Wash Buffer gently.
34. Determine the concentration of the cell suspension using cell counting equipment. Using a 0.5 mL lobind tube, add 18 µL of ice-cold PBS and 2 µL of the nuclei suspension. Pipette mix and transfer 9 µL of the diluted nuclei to a new 0.5 mL tube. Add 1uL of AOPI dye. Mix and load 10 µL in the Luna slide.
35. Record nuclei concentration and calculate desired suspensions to freeze which require a concentration between 3,000-10,000 cells/µL.

Example: *(If desired concentration is 10,000 cell/uL)*

Current Concentration (C_c) = 1.2x1^6 cells/mL
Target Concertation (C_t)= 10,000 cells/uL
Dilution factor (DF) = 1:10
Current volume (V_c) = 80uL = 0.08mL
Total Volume Needed = $V = \frac{C_c \times DF \times V_c}{C_t} = \frac{1.2x10^6 cells/mL \times 10 \times 0.08mL}{10{,}000\ cells} = \mathbf{96\ \mu L}$

Now Subtract the Total Volume needed minus the current volume, to obtain the volume needed to achieve the desired concentration.

$$V = 96 - 80 = \mathbf{16\ \mu L}$$

36. Store the fixed nuclei in a pre-labeled Scale 1.5 Eppendorf, place tube in the Scale Box in the -80°C freezer and record the scale poll ID.